

Parallel and distributed Computing

Lecture 10

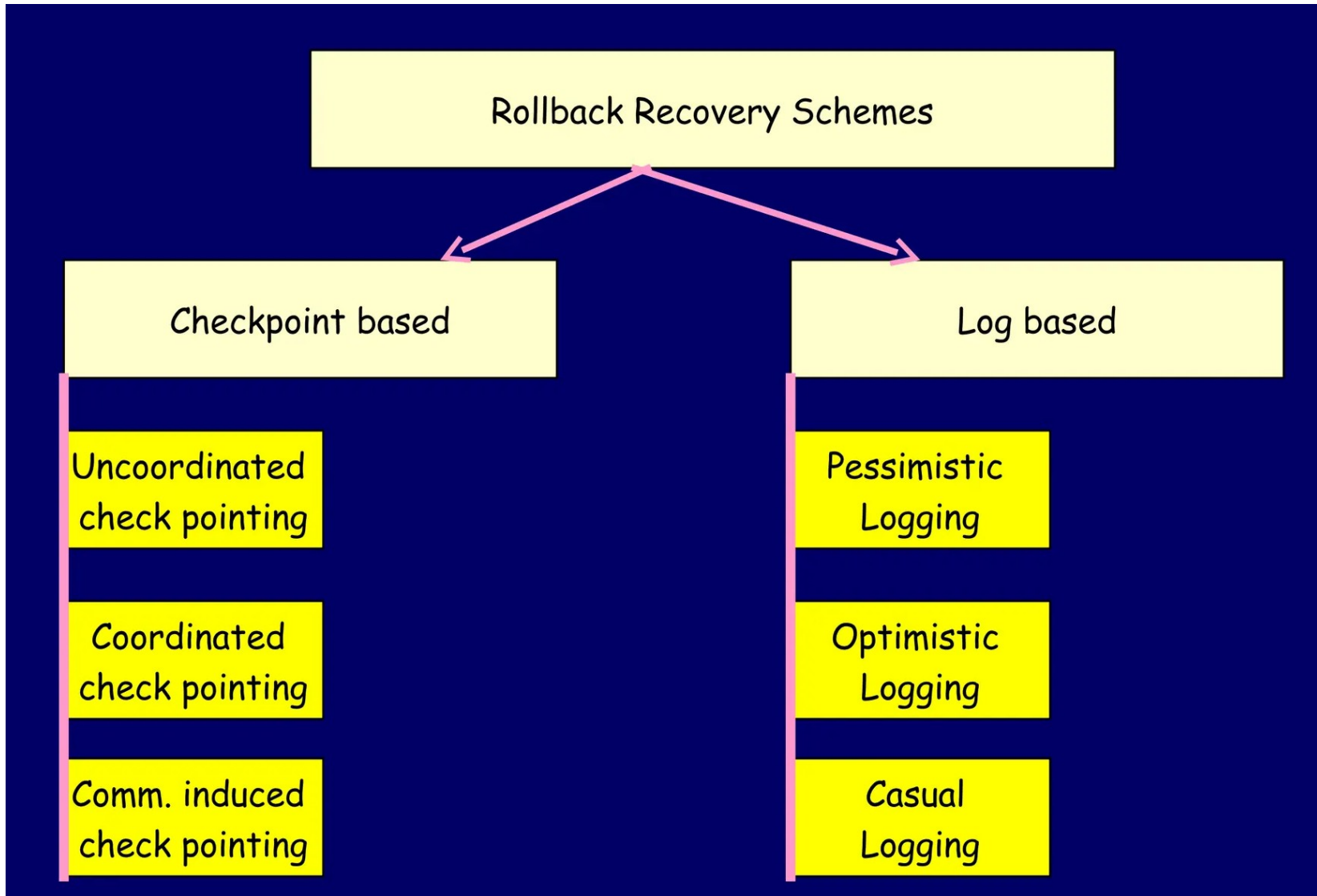
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- Rollback recovery treats a distributed system as a collection of processes that communicate through a network
- Fault tolerance is achieved by periodically using stable storage to save the processes' states during the failure-free execution.
- Upon a failure, a failed process restarts from one of its saved states, thereby reducing the amount of lost computation.
- Each of the saved states is called a checkpoint

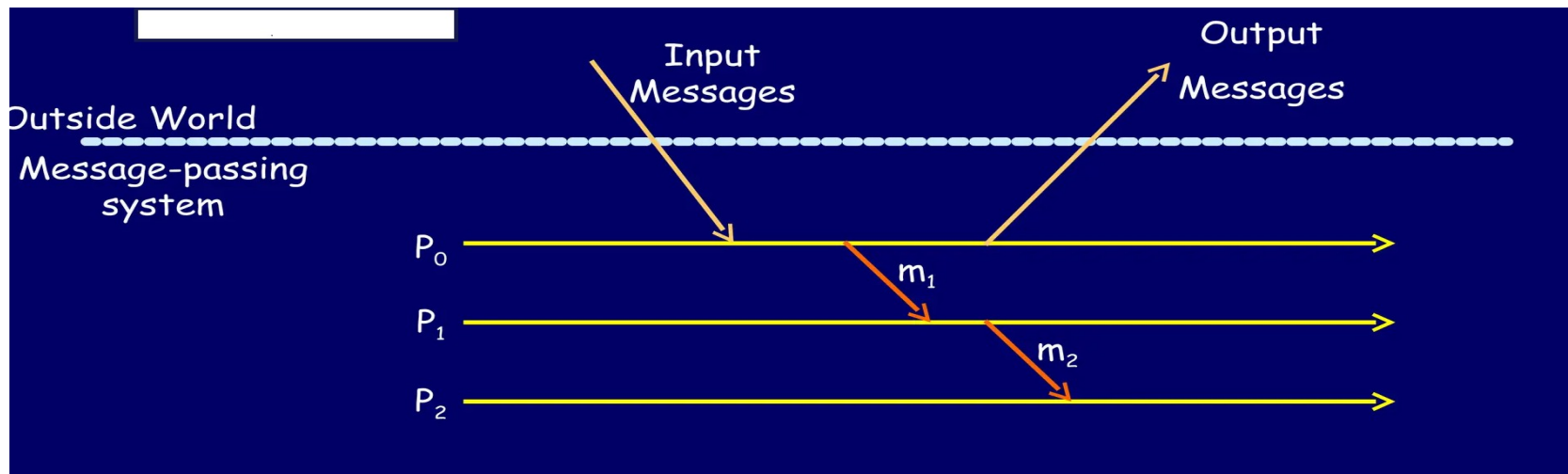
Different Rollback Recovery schemes



Checkpoint based Recovery

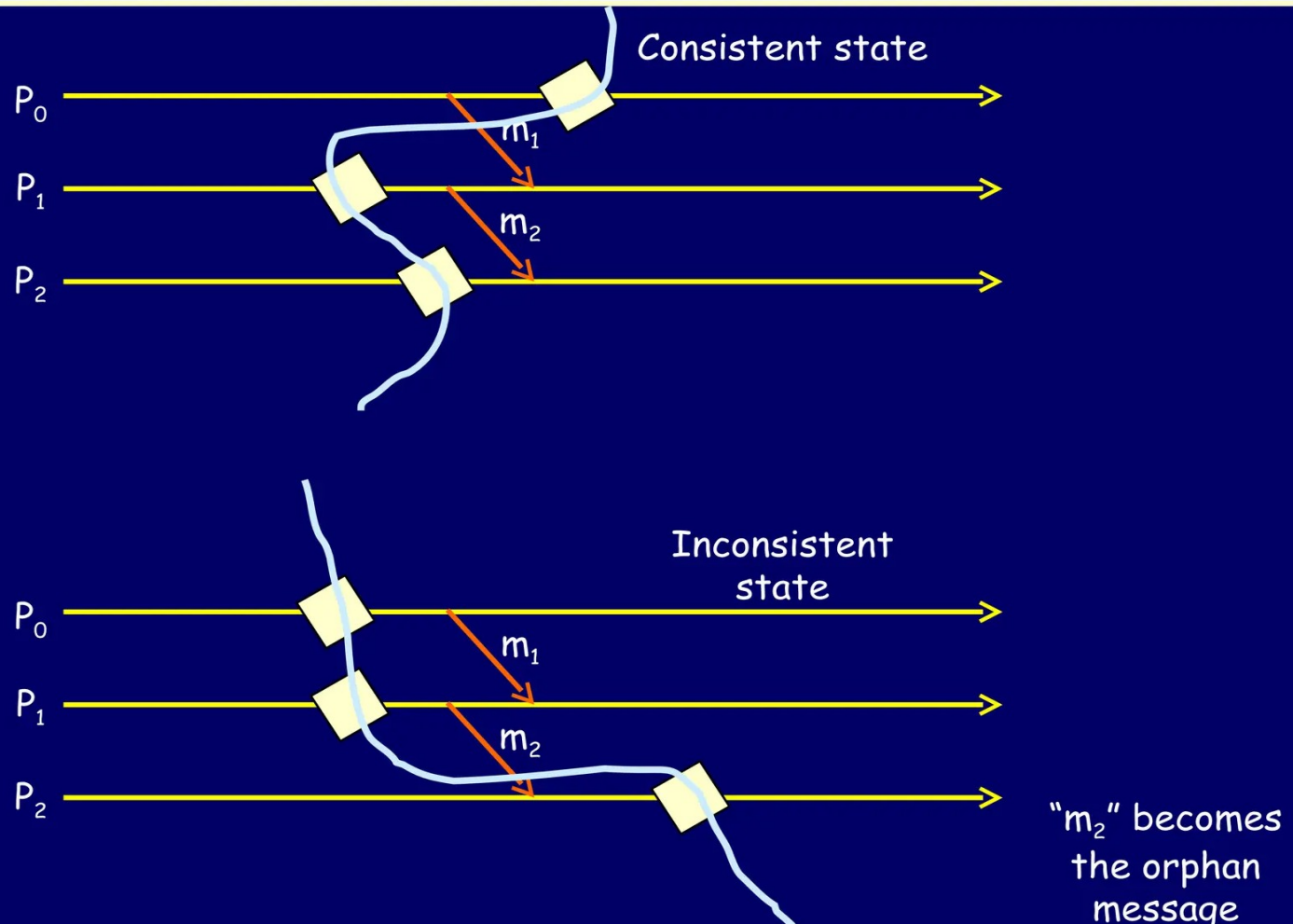
- **Uncoordinated checkpointing:** Each process takes its checkpoints independently
- **Coordinated checkpointing:** Process coordinate their checkpoints in order to save a system-wide consistent state. This consistent set of checkpoints can be used to bound the rollback
- **Communication-induced checkpointing:** It forces each process to take checkpoints based on information hold on the application messages it receives from other processes.

- System consist of a fixed number (N) of processes which communicate only through messages
- Processes cooperate to execute a distributed application program and interact with outside world by receiving and sending input and output messages respectively.



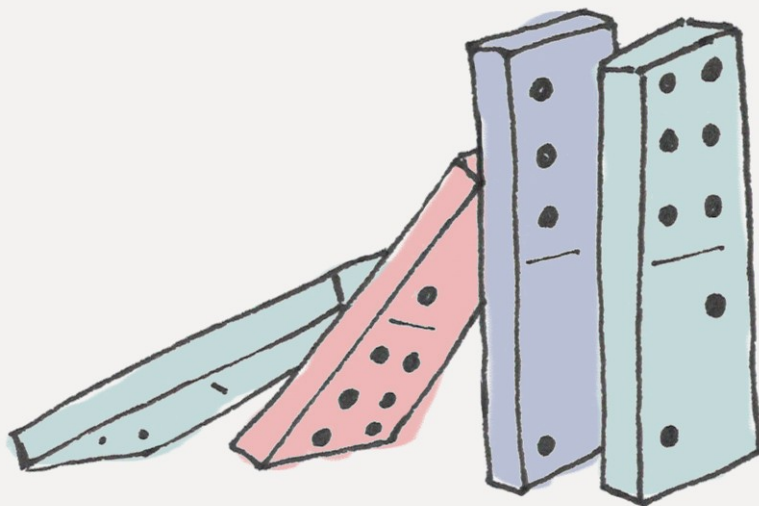
- A consistent system state is one in which if a process's state reflects a message receipt, then the state of the corresponding sender reflects sending that message.
- A fundamental goal of any rollback-recovery protocol is to bring the system into a consistent state when inconsistencies occur because of a failure.

Example



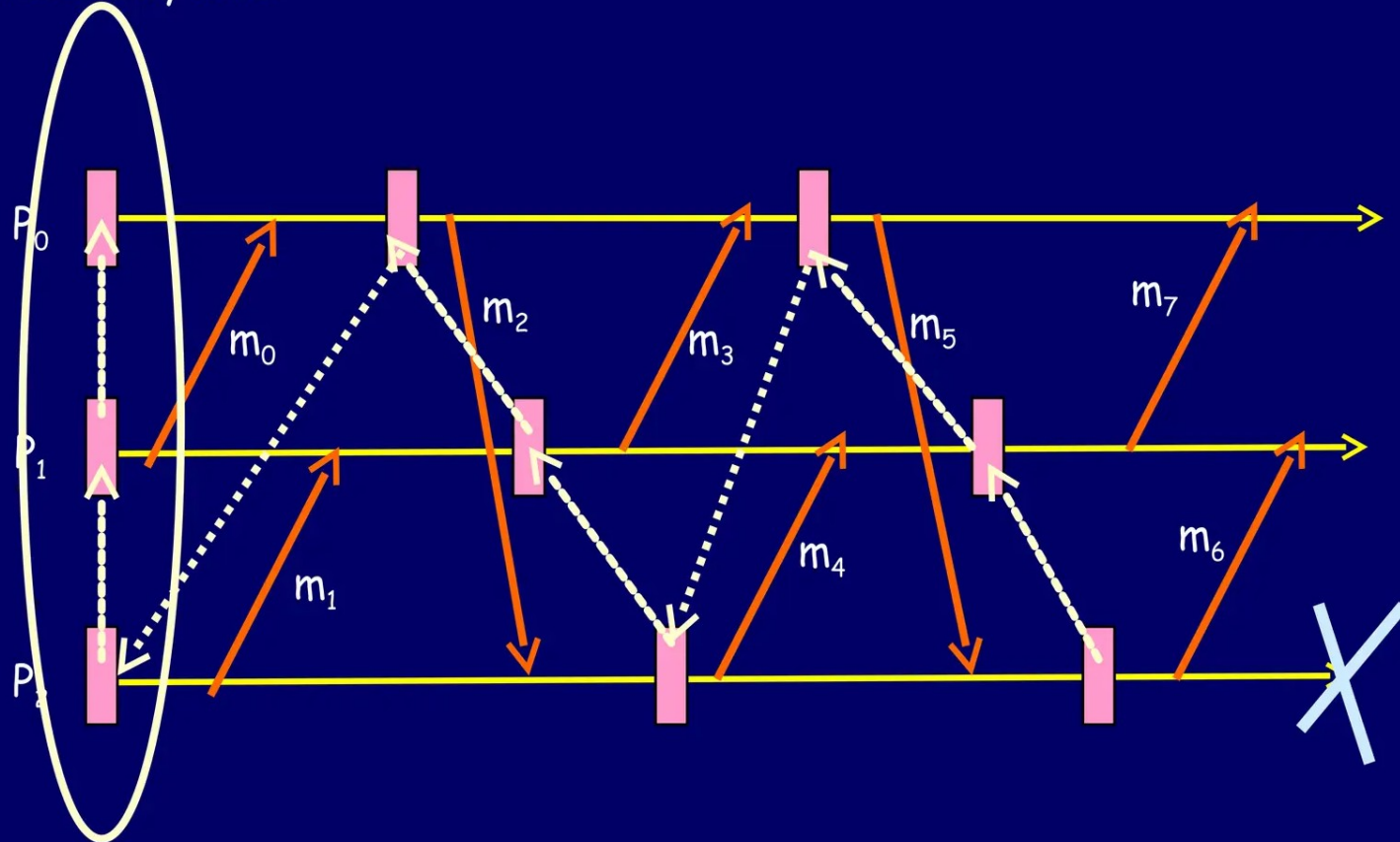
- Each process periodically saves its state on stable storage. The saved state contains sufficient information to restart process execution.
- A consistent global checkpoint is a set of N local checkpoints, one from each process, forming a consistent system state.
- Any consistent global checkpoint can be used to restart process execution upon a failure.
- The most recent consistent global checkpoint is termed as the recovery line.
- In the uncoordinated checkpointing paradigm, the search for a consistent state might lead to **domino effect**.

Domino effect



Example

Recovery Line



Domino Effect: Cascaded rollback which causes the system to roll back to too far in the computation (even to the beginning), in spite of all the checkpoints

- A message passing system often interacts with the outside world to receive input data or show the outcome of a computation. If a failure occurs the outside world cannot be relied on to rollback.
- For example, a printer cannot rollback the effects of printing a character, and an automatic teller machine cannot recover the money that it dispensed to a customer.
- It is therefore necessary that the outside world perceive a consistent behavior of the system despite failures.

- Thus, before sending output to the outside world, the system must ensure that the state from which the output is sent will be recovered despite of any future failure
- Similarly, input messages from the outside world may not be regenerated, thus the recovery protocols must arrange to save these input messages so that they can be retrieved when needed.

- Checkpoints and event logs consume storage resources.
- As the application progresses and more recovery information collected, a subset of the stored information may become useless for recovery.
- Garbage collection is the deletion of such useless recovery information.
- A common approach to garbage collection is to identify the recovery line and discard all information relating to events that occurred before that line.

- **Uncoordinated Check pointing**
- Allows each process maximum independency in deciding when to take checkpoints
- **Advantage:** each process may take a checkpoint when it is most convenient
- **Disadvantages:**
 - Domino effect
 - Possible useless checkpoints
 - Need to maintain multiple checkpoints
 - Garbage collection is needed
 - Not suitable for applications with outside world interaction (output commit)

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.